

1) Define Depreciation and explain different types of depreciation With example.

* The Recovery of money from the earnings of an equipment for its replacement purpose is called depreciation fund since we make an assumption that the value of the equipment decrease with the passage of time.

* Thus, the word "depreciation" means decrease in value of any physical asset with passage of time.

Types of Depreciation:

- * Straight line method of Depreciation
- * Declining balance method of depreciation
- * Sum of years - digits method of depreciation
- * Sinking fund method of depreciation
- * Service output method of depreciation

* Straight line method of depreciation:

* In this method of depreciation, a fixed sum is charged as the depreciation amount throughout the lifetime of an asset such that the accumulated sum at end of life of asset is exactly equal to the purchase value of asset. Here, we make an important assumption that inflation is absent.

Let ,
 P = first cost of asset ,
 F = salvage value of asset ,
 n = life of asset ,
 B_t = Book value of asset at end of period t .
 D_t = Depreciation amount of period t .

Depreciation and Book value are,

$$D_t = (P - F) / n$$

$$B_t = B_{t-1} - D_t = P - t \times [(P - F) / n]$$

Example:- A company has purchased a equipment whose first cost is Rs. 1,00,000 with an estimated life of eight years. The estimated salvage first value of equipment at end of its lifetime is Rs. 20,000. Determine the depreciation charge and book value of end of various years using the straight line method of Depreciation.

$$\Rightarrow P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$D_t = (P - F) / n = (1,00,000 - 20,000) / 8 \\ = \text{Rs. } 10,000.$$

End of years (t)	Depreciation (D_t)	Book value ($B_t = B_{t-1} - D_t$)
0	-	1,00,000
1	10,000	90,000
2	10,000	80,000
3	10,000	70,000
4	10,000	60,000
5	10,000	50,000
6	10,000	40,000
7	10,000	30,000
8	10,000	20,000

In this method of depreciation, the value of D_t is the same for all the years.

* Declining Balance method of Depreciation.

* In this method of depreciation, a constant percentage of book value of previous period of the asset will be charged as depreciation amount of current period.

* This Approach is a more realistic approach, since the depreciation charge decreases with the life of asset which matches with the earning potential of asset.

Let,
 P = First cost of asset,
 F = salvage value of asset,
 n = life of asset.
 B_t = Book value of asset at end of period t .
 K = a Fixed percentage
 D_t = depreciation amount at the end of period t .

The formula of depreciation and Book value,

$$D_t = K \times B_{t-1}$$

$$B_t = B_{t-1} - D_t = B_{t-1} - K \times B_{t-1} \\ = (1-K) \times B_{t-1}$$

In terms of P ,

$$D_t = K(1-K)^{t-1} \times P$$

$$B_t = (1-K)^t \times P$$

Example:- A company has purchased an equipment whose first cost is Rs. 1,00,000 with an estimate life of eight years. The estimated Salvage value of equipment at end of its lifetime is Rs. 20,000. Determine the depreciation charge and book value at end of various years using the ~~straight~~ declining balance method

$$\Rightarrow P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$K = 0.2.$$

The calculations pertaining to D_t and B_t for different values of t .

$$D_t = K \times B_{t-1}$$

$$B_t = B_{t-1} - D_t$$

End of years (n)	Depreciation (Dt)	Book value (Bt)
0	20,000.00	1,00,000
1	16,000	80,000
2	12,800	64,000
3	10,240	51,200
4	8,192	40,960
5	6,553.60	32,768
6	5,242.88	26,214.40
7	4,194.30	20,971.52
8		16,777.22

*) Sum of the years - digits method of depreciation.

* In this method of depreciation, it is assumed that the book value of the asset decrease at a decreasing rate. If the asset has a life of eight years, first the sum of years is computed as

$$\text{Sum of years} = 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36 = n(n+1)/2$$

The Rate of depreciation charge for first year is assumed as highest and then it decreases. The Rate of depreciation for years 1-8 respectively as follows: $8/36, 7/36, 6/36, 5/36, 4/36, 3/36, 2/36$ & $1/36$. For any year, the depreciation with (P-F)

$$Dt = \text{Rate} \times (P-F)$$

$$Bt = B_{t-1} - Dt$$

$$\Rightarrow Dt = \frac{n-t+1}{n(n+1)/2} (P-F)$$

$$Bt = (P-F) \frac{(n-t)(n-t+1)}{2(n+1)} + F$$

Example:- A company has purchased an equipment whose cost is Rs. 1,00,000 with an estimated life of 8 years. The estimated salvage value of equipment at end of lifetime is Rs. 20,000. Determine the depreciation charge and Book value.

$$P = \text{Rs. } 1,00,000$$

$$F = \text{Rs. } 20,000$$

$$n = 8 \text{ years}$$

$$\text{Sum} = n(n+1)/2 = 8 \times 9 / 2 = 36$$

$$Dt = \text{Rate} \times (P-F)$$

$$Bt = Bt-1 - Dt$$

⇒ Dt and Bt under Sum of years-digits method of depreciation

End of year (n)	Depreciation (Dt)	Book value (Bt)
0	-	1,00,000
1	17,777.77	82,222.23
2	18,555.55	66,666.68
3	13,333.33	53,333.35
4	11,111.11	42,222.24
5	8,888.88	33,333.36
6	6,666.66	26,666.70
7	4,444.44	22,222.26
8	2,222.22	20,000.04

* Sinking Fund method of depreciation.

* In this method, the Book value decrease at increasing rates with respect to life of asset. Let,

P = First cost of asset,

F = Salvage value,

n = life of asset

i = Rate of (interest) return compounded Annually.

A = the annual equivalent Amount,

Bt = Book value

Dt = the depreciation Amount.

The depreciation Amount (A) at the end of each Period during the lifetime.

$$A = (P-F) \times (A/F, i, n)$$

$$\Rightarrow Dt = (P-F) \times (A/F, i, n) \times (F/P, i, t-1)$$

$$Bt = P - (P-F) (A/F, i, n) (F/A, i, t)$$

Example:- A company has purchased an equipment whose cost is Rs. 1,00,000 with estimate life of 8 years. The estimated salvage value of equipment at end of lifetime is Rs. 20,000. and compute D_t and B_t using Sinking Fund method of depreciation with an interest rate of 12% compounded annually.

$$\Rightarrow \begin{aligned} P &= \text{Rs. } 1,00,000 \\ F &= \text{Rs. } 20,000 \\ n &= 8 \text{ years} \\ i &= 12\% \end{aligned}$$

$$\begin{aligned} D_t &= (P-F) (A/F, i, n) (F/F, i, t-1) \\ &= (P-F) (A/F, 12\%, 8) (F/P, 12\%, 4) \\ &= (1,00,000 - 20,000) \times 0.0813 \times 1.574 \\ &= \text{Rs. } 10,237.30 \end{aligned}$$

$$\begin{aligned} B_t &= P - (P-F) (A/F, i, n) (F/A, i, t) \\ &= P - (P-F) (A/F, 12\%, 8) (F/A, 12\%, 4) \\ &= 1,00,000 - (1,00,000 - 20,000) \times 0.0813 \times 10.089 \\ &= 34,381.10 \end{aligned}$$

* Service output method of depreciation:

* In some situations, it may not be realistic to compute depreciation based on time period. In such cases, the depreciation is computed based on service rendered by an asset. Let,

P = First cost of asset

F = Salvage value

X = maximum capacity of service

x = quantity of service rendered in period

$$\text{Depreciation / unit of service} = (P-F) / X$$

Depreciation for x unit of service in

$$\text{a period} = \frac{P-F}{X} (x)$$

Example: The First cost of road laying machine is Rs. 80,00,000. Its Salvage value after five years is Rs. 50,000. The length of road that can be laid by machine during its lifetime is 75,000 km. In its third year of operation, the length of road laid is 2,000 km. Find the depreciation of equipment for that year.

$$\Rightarrow \begin{aligned} P &= \text{Rs. } 80,000,000 \\ F &= \text{Rs. } 50,000 \\ X &= 75,000 \text{ km} \\ x &= 2,000 \text{ km} \end{aligned}$$

Depreciation for x units of service in period

$$\begin{aligned} &= \frac{P-F}{X} x \\ \text{depreciation for year 3} &= \frac{(80,00,000 - 50,000)}{75,000} \times 2,000 \\ &= \text{Rs. } 2,12,000. \end{aligned}$$